# Flexibility vs. Interpretability in High-Dimensional Cancer Classification

Regularized Regression, Tree Ensembles, and Neural Networks

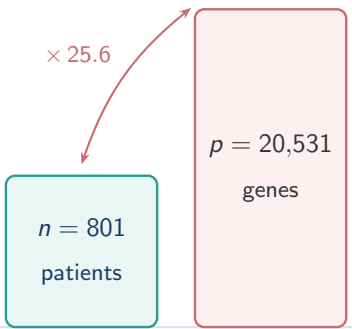
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Regression Techniques Course Project · Indian Statistical Institute

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# Why Is High-Dimensional Gene Expression Hard?






**Statistical bottlenecks in $p \gg n$:**

- **Singular covariance:** $\mathrm{rank}(\mathbf{S}) \leq 800 \ll 20{,}531$; classical LDA fails.
- **Severe multicollinearity:** co-expressed genes ($|r| > 0.85$) destabilize selection.
- **High-variance estimation:** unconstrained models overfit noise without shrinkage.

▷ **Regularization is essential for well-posed estimation.**

SECTION 01

# Dataset and Exploratory Analysis

# TCGA RNA-Seq Dataset

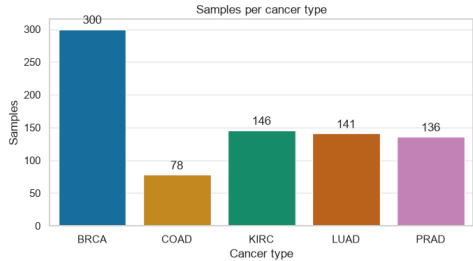


**801** samples
Total patients

**5** classes
BRCA KIRC COAD
LUAD PRAD

**1,000** genes
MAD pre-screen

**20,531** genes
Raw transcriptome



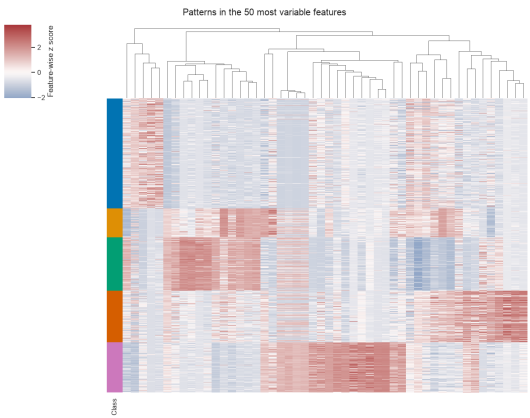

**Preprocessing pipeline:**

- **Log-transform:** $z_{ij} = \log_2(x_{ij} + 1)$ stabilizes expression variance.
- **Feature screening:** Top-1,000 genes by in-fold MAD (zero test leakage).
- **Standardization:** In-fold $Z$-score scaling ($\mu = 0$, $\sigma = 1$) per feature.

# Expression Heatmap: 50 Highest-Variance Genes



Unsupervised hierarchical clustering reveals tissue-specific expression blocks




**Key observations:**

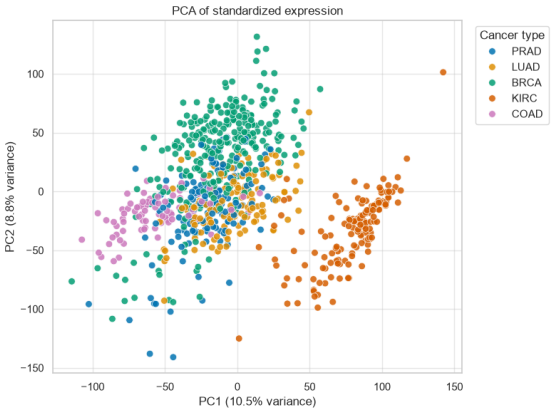
- Colours show standardised expression per gene; columns are patients.
- Cancer cohorts form **distinct blocks** without any labels.
- Row clusters reflect shared **gene modules** and pathways.
- Strong visual separation motivates linear classification.

▷ **The dataset carries strong class-specific structure even in unsupervised clustering.**

# PCA: Tumours Separate in Two Linear Dimensions



Spectral projection of 1,000 variance-filtered genes onto the leading principal components




- **PC1 (18.6%):** Separates BRCA from KIRC.
- **PC2 (12.3%):** Separates COAD and LUAD.
- **30.9%** of total variation visible in 2D.
- Five cohorts are almost fully linearly separable in this unsupervised projection.

▷ **Linear separability exists before any model fitting — non-linear methods gain little here.**

SECTION 02

# Classification Models

# Multinomial Logistic Regression & Regularization

Model class probabilities using the softmax function:

$$P(Y_i = k \mid x_i) = \frac{\exp(x_i^\top \beta_k)}{\sum_{j=1}^{5} \exp(x_i^\top \beta_j)}$$

Estimate coefficients by minimizing multiclass cross-entropy with a regularization penalty:

$$\mathcal{L}(B) = \mathcal{L}_{\text{CE}}(B) + \lambda P(B)$$

## 3. Regularization strategies

| Model | Penalty | Statistical effect |
|---|---|---|
| **Ridge** | $\frac{1}{2}\|B\|_F^2$ | Shrinks coefficients |
| **Lasso** | $\|B\|_1$ | Produces sparse coefficients |
| **Elastic Net** | $\alpha\|B\|_1 + \frac{1-\alpha}{2}\|B\|_F^2$ | Combines shrinkage and sparsity |

# Non-Linear Classifiers



### 1. Random Forest

- Ensemble of bootstrap-trained trees.
- Random feature subsets at each split.
- Captures nonlinearities and feature interactions.

### 2. XGBoost

- Gradient-boosted decision trees.
- Sequentially corrects previous prediction errors.
- Captures complex nonlinear decision boundaries.

### 3. PCA-MLP

- PCA reduces the dimensionality of gene expression.
- Multilayer perceptron learns nonlinear mappings.
- Combines dimensionality reduction with neural modelling.

### 4. Sparse Group-Lasso NN

- Neural network with group-based sparsity.
- Encourages selective use of input features.
- Combines nonlinear learning with feature regularization.

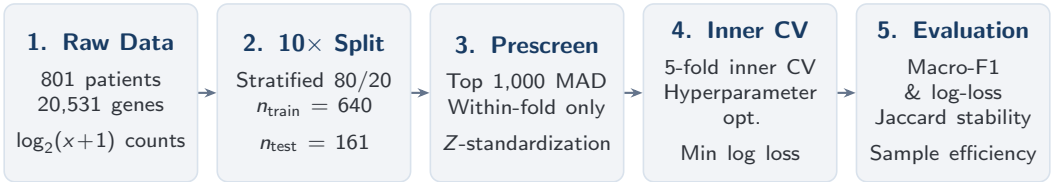
SECTION 03

# Experimental Design

# Zero-Leakage Evaluation Pipeline



10 repeated stratified splits with strictly within-fold feature prescreening




▷ **Strict zero-leakage: feature screening and standardization are isolated within training folds.**

# Why Three Metrics?



Accuracy, probability calibration, and biomarker stability

## Macro-F1 (Accuracy)

$$\frac{1}{5}\sum_{k=1}^{5}\frac{2P_kR_k}{P_k+R_k}$$

- Harmonic mean across classes.
- **Ceiling:** all models $> 0.99$.

## Log-Loss (Calibration)

$$-\frac{1}{N}\sum_{i,k} y_{ik}\log\hat{p}_{ik}$$

- Penalizes overconfidence.
- **Decisive metric** for ranking.

## Stability (Biomarkers)

$$J = \frac{|S_a \cap S_b|}{|S_a \cup S_b|}$$

- Gene overlap across splits.
- **Key** for clinical biomarkers.

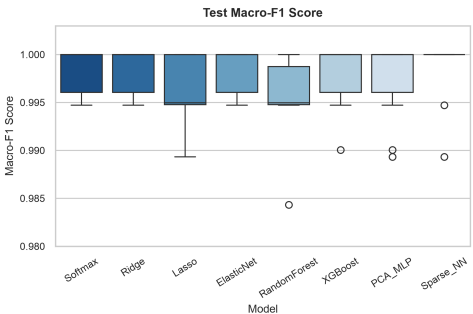
SECTION 04

# Results

# All Models Achieve High Accuracy — Log-Loss Tells the Real Story



Performance distributions across 10 repeated train/test splits




**F1 saturates** above 0.993 for all models; accuracy alone cannot distinguish architectures.




**Log-loss diverges:** tree models (0.081) vs. regularized linear models ($< 0.020$).

# Comprehensive Model Comparison



10-split cross-validated test results

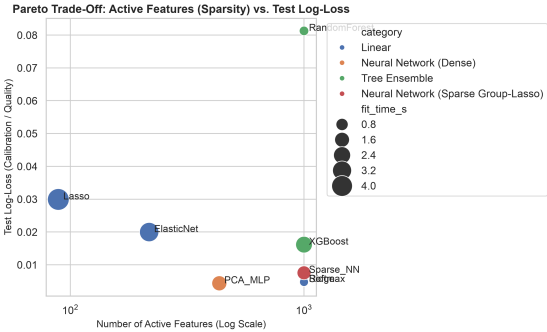
| Model | Macro-F1 | Log-Loss | Genes | Jaccard |
|---|---|---|---|---|
| Softmax | 0.998 | 0.005 | 1000 | — |
| Ridge | 0.998 | 0.005 | 1000 | — |
| Lasso | 0.996 | 0.030 | **89** | 0.740 |
| **Elastic Net** | **0.998** | **0.020** | **218** | **0.773** |
| Random Forest | 0.995 | 0.081 | 1000 | 0.242 |
| XGBoost | 0.998 | 0.016 | 1000 | 0.467 |
| PCA-MLP | 0.997 | 0.004 | 434 | — |
| Sparse NN | 0.998 | 0.008 | 1000 | — |

▷ **Elastic Net matches top F1 with 218 genes and leads sparse stability.**

# Sparsity vs. Calibration: The Pareto Frontier



Elastic Net achieves optimal balance between features and calibration



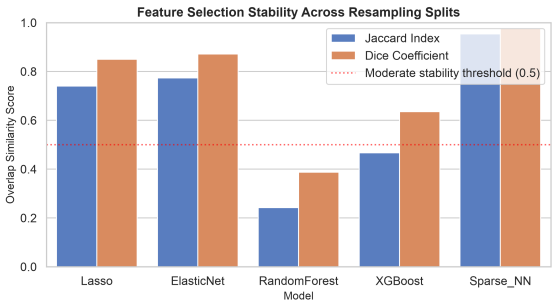

- **Ridge / PCA-MLP:** Dense (1000 genes / 434 PCs), log-loss 0.004–0.005.
- **Lasso:** Highly sparse (89 genes), log-loss 0.030.
- **Elastic Net:** Optimal trade-off (218 genes, log-loss 0.020).
- **Tree models:** Off frontier (log-loss 0.081).

▷ **Elastic Net occupies the optimal sparse Pareto frontier.**

# Feature Selection Stability



Selection consistency across 10 independent data splits



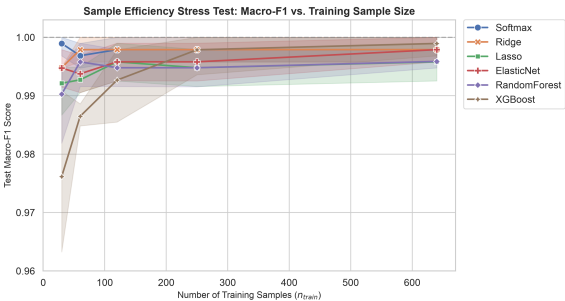

- **Elastic Net:** Jaccard 0.77 (Dice 0.87) — leads sparse models.
- **Lasso:** Jaccard 0.74 — collinear omission hurts stability.
- **XGBoost:** Jaccard 0.47 — greedy splits fluctuate.
- **Random Forest:** Jaccard **0.24** — lowest consistency.

▷ **$L_2$ grouping is essential for reproducible biomarker discovery.**

# Sample Efficiency Under Data Scarcity



Macro-F1 as training size drops from 640 to 30




- **Ridge & Elastic Net:** Maintain $F_1 \geq 0.995$ down to $n_{\text{train}} = 30$.
- **XGBoost:** Drops to 0.976 under data scarcity.
- **Random Forest:** Degrades to 0.985 at $n = 30$.

Y-axis starts at 0.96 to highlight low-$n$ divergence.

▷ **Regularized linear models remain sample-efficient; trees degrade sharply.**

SECTION 05

# Interpretation and Conclusions

# Three Main Findings



Answering the three primary research questions of this study

**Q1** **Flexibility provides no advantage in RNA-Seq tumour classification.**

Linear separability holds in gene space. Complex non-linear models yield zero F1 gain and worse calibration.

**Q2** **High collinearity severely destabilises Lasso and decision trees.**

Lasso selects arbitrarily among correlated genes; tree stability drops to 0.24. Elastic Net grouping recovers stability (0.77).

**Q3** **Elastic Net achieves the optimal sparsity–stability–calibration trade-off.**

Ties top Macro-F1 (0.998), eliminates 78% of candidate genes (218 retained), and delivers highest sparse biomarker consistency.

# Limitations



### 1. Single-Cohort Evaluation

Analysis uses 801 samples from one dataset; generalizability to independent cohorts is unverified.

### 2. High-Dimensional Feature Space

Screening 20,531 genes to 1,000 potentially excludes relevant biological features.

### 3. Limited Biological Interpretability

Anonymized gene identifiers restrict biological interpretation of selected features.

### 4. Broad Cancer Classification

Distinguishes five cancer types without examining molecular subtypes within each type.

# Thank You